Ahmedabad University

Weekly Report 2

Disease Prediction System using Machine Learning

Submitted to faculty: Mehul Raval

Student: Ayushi Patel

Enrollment No. : AU2519010

Course: CSE 620 Machine Learning

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Week 2 Dataset Exploration and Structural Analysis

Work Completed

This week, the selected symptom-disease dataset was imported into the Python environment for preliminary analysis. The dataset structure was carefully reviewed to understand its composition, including the total number of records, the number of symptom-related feature columns, and the corresponding disease labels.

An exploratory analysis was conducted to examine how symptoms are represented in the dataset. Sample records were inspected to verify that symptom attributes are consistently organised and suitable for use as input features in a machine learning model. The dataset was analysed for possible issues such as missing values, duplicate records, and inconsistent data formats that could affect the training process.

Basic statistical observations were also conducted to gain an overview of the dataset and to understand the distribution of different disease classes. This step helped assess whether the dataset provides sufficient representation of diseases to train a reliable predictive model.

Tools / Technologies

- Python
- Pandas

Challenges

Understanding the symptom representation format and ensuring that it can be effectively converted into numerical features for model training.

Plan for Next Week

- Perform preprocessing of symptom features.
- Prepare the dataset for machine learning training.